

- 2 (a) weight = 452×9.81
 component down the slope = $452 \times 9.81 \times \sin 14^\circ$
 = $1072.7 = 1070 \text{ N}$ M1
 A0 [1]
- (b) (i) $F = ma$ C1
 $T - (1070 + 525) = 452 \times 0.13$ C1
 $T = 1650 \text{ (1653.76) N}$ any forces missing 1/3 A1 [3]
- (ii) 1. $s = ut + \frac{1}{2}at^2$ hence $10 = 0 + \frac{1}{2} \times 0.13t^2$ C1
 $t = [(2 \times 10) / 0.13]^{1/2} = 12.4 \text{ or } 12 \text{ s}$ A1 [2]
2. $v = (0 + 2 \times 0.13 \times 10)^{1/2} = 1.61 \text{ or } 1.6 \text{ m s}^{-1}$ A1 [1]
- (c) straight line from the origin B1
 line down to zero velocity in short time compared to stage 1 B1
 line less steep negative gradient B1
 final velocity larger than final velocity in the first part – at least $2\times$ B1 [4]